Concept map: Foundations and framing

Taylor series
XGBoost
Memoization
Moore-Penrose pseudo-inverse
Derivatives of one variable
Backpropagation
Linear algebra roadmap
Latent semantic analysis
Determinant
Normal equation
Gradient descent
Low-rank approximation (truncated SVD)
Multicollinearity
Hessian and multivariate Taylor
Eigenvectors and eigenvalues
Forward propagation
Universal approximation theorem
Singular value decomposition
Jacobian and matrix gradients
Partial derivatives and gradients
Linear transformations and matrices
Linear combinations, span and basis
Convex and non-convex loss
Ridge regression
Lasso regression
Multi-layer perceptron (MLP)
PCA
Matrix multiplication as composition
Perceptron
Perceptron trick
History of deep learning
Neural networks
Cosine similarity
Lagrange multipliers, KKT and duality
Bag of words
Regularisation
Anomaly detection
Dot product
Convex sets and convex optimisation
Clustering
Vectors and feature vectors
Equation of a hyperplane
Dimensionality reduction
Unsupervised learning
Vector magnitude, distance and scalar operations
Linear and quadratic programming
Association rule learning
Deep learning
Support vector machines
Pareto distribution and power laws
Hyperparameter tuning
Function transformer
Tensors
Logistic regression
Applications of ML
Reinforcement learning
K-nearest neighbours
Decision trees
How to learn the maths for ML
Role of mathematics in ML
Machine learning
Instance-based learning
Log-normal distribution
Voting ensembles
Data mining
Model-based learning
Uniform distribution
Discrete and continuous data
Exploratory data analysis
Symbolic AI and expert systems
Semi-supervised learning
What deep learning is
Classification problems
Bagging
Types of neural networks
Features
Supervised learning
Independent and mutually exclusive events
Bernoulli and binomial distributions
Random variables
Artificial intelligence
Simple linear regression
Accuracy
Expected value and variance of a random variable
Feature engineering
Setup: conda, Jupyter and Colab
Decision surface and boundary
Probability distributions
Measures of central tendency
Z-score outlier method
Representation learning
Labelled data
Naive Bayes
Probability density function (PDF)
Poisson distribution
Pandas Profiling
Regression problems
Q-Q plot
Probability mass function (PMF)
Framing an ML problem
Normal distribution
Regression metrics
Standard normal and the z-table
Cumulative distribution function (CDF)
Batch (offline) learning
Bayes' theorem
Multiple linear regression
Central limit theorem
Student's t-distribution
ML development life cycle
Retraining
Z-test and rejection regions
Online learning
Correlation significance test
T-tests: one-sample, two-sample, paired
Choosing a hypothesis test
Model drift
Conditional probability
Joint and marginal probability
Confidence intervals
P-values
Out-of-core learning
Sampling distribution and standard error
Stochastic gradient descent
Bessel's correction
Hypothesis testing, null and alternative
Mini-batch gradient descent
Chi-square tests
One-sample proportion test
Venn diagrams and contingency tables
One-way ANOVA
Events and sample spaces
Inferential statistics
Type I and II errors, power, tails
Population, sample, parameter and statistic
Beta and A/B testing
Feature selection
Variance
Confusion matrix
Empirical vs theoretical probability, probability rules
Descriptive statistics
Sampling noise and bias
Density estimation

Links: needs | is a kind of | is a kind of (draft) | fixes | compared with | used in | used in (draft)